

Self-condensation of Terminal Acetylenic Ketones via Organozinc Species

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A self-condensation reaction of α,β -acetylenic ketones via organozinc species $\text{CF}_3\text{CO}_2\text{ZnEt}$ to afford the β -branched Morita-Baylis-Hillman adducts has been developed. This reaction system provides functionalized trisubstituted olefins in good yields. Organozinc species $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$, an efficient cyclopropanation reagent, can also react with aromatic acetylenic ketones to provide 2,5-dihydrofurans in moderate yields.

Keywords acetylenic ketone, organozinc species, trisubstituted olefin, 2,5-dihydrofuran

Introduction

The Morita-Baylis-Hillman reaction is an efficient carbon-carbon bond-forming reaction in organic synthesis.^{1,2} However, the β -substituted olefinic substrates do not normally undergo the Baylis-Hillman reaction.³ In order to overcome the limitation of the original Morita-Baylis-Hillman reaction, alternative methods, such as the reaction of copper,⁴ aluminum^{5,6} and boron reagents⁷ with α,β -acetylenic carbonyl compounds have been developed.

Recently, we reported a new procedure for the stereoselective synthesis of β -branched Morita-Baylis-Hillman adducts by a one-pot, three-component reaction of α,β -acetylenic ketones with organozinc species and aldehydes.⁸ In the process of this coupling reaction, small amounts (<5%) of self-condensation products from acetylenic ketones were also obtained as byproducts. When the reaction was carried out in the absence of aldehydes, the self-condensation products were obtained in good yields. Herein, we wish to report this self-condensation reaction of α,β -acetylenic ketones with organozinc species $\text{CF}_3\text{CO}_2\text{ZnEt}$ to afford unusual β -branched Morita-Baylis-Hillman adducts.

Experimental

All reactions were conducted under nitrogen atmosphere in oven-dried glassware with magnetic stirring. Dichloromethane was dried and freshly distilled from calcium hydride under nitrogen atmosphere. Chromatographic purification was performed on silica gel (100–200 mesh) and analytical thin layer chromatography (TLC) on silica gel 60-F₂₅₄, which was detected by fluorescence. ¹H NMR (300 MHz) and ¹³C NMR (75 MHz) spectra were measured with a Bruker AC 300

spectrometer using TMS as an internal standard. High resolution mass spectra were obtained with a Micromass GCT-TOF mass spectrometer. IR spectra were recorded as thin films or as solids in KBr pellets on a Perkin-Elmer FT210 spectrophotometer.

General procedure

A 50 mL round-bottomed flask was equipped with a stir bar and charged with freshly distilled methylene chloride (4 mL), and diethylzinc (70 μL , 0.69 mmol) was added via a syringe under nitrogen. The solution was cooled in an ice bath and trifluoroacetic acid (54 μL , 0.69 mmol) was then dropped very slowly into the reaction mixture via a syringe. Upon stirring for 30 min, CH_2I_2 (58 μL , 0.69 mmol) was added dropwise via a syringe under nitrogen. After an additional 30 min stirring at 0 °C, the α,β -acetylenic ketone (1.0 mmol) was added, and the ice bath was removed. The reaction mixture was stirred for 24 h at room temperature. The solution was quenched by saturated aqueous ammonium chloride solution and extracted with diethyl ether (10 $\times$ 3 mL). The combined organic layers were washed with brine, then dried over anhydrous magnesium sulfate, filtered, and concentrated on rotary evaporator. The desired product was isolated by silica gel chromatography with petroleum ether/EtOAc (10 : 1–5 : 1).

(Z)-3-(2-Hydroxybut-3-yn-2-yl)hex-3-en-2-one 2a:

The desired product was obtained according to the general procedure without addition of CH_2I_2 ; A yellow oil (72% yield); ¹H NMR (CDCl_3 , 300 MHz) δ : 5.97 (t, J = 7.7 Hz, 1H), 3.34 (s, 1H), 2.52 (s, 1H), 2.36 (s, 3H), 2.10 (dq, J = 7.7, 7.4 Hz, 2H), 1.60 (s, 3H), 1.02 (t, J = 7.4 Hz, 3H); ¹³C NMR (CDCl_3 , 75 MHz) δ : 206.8, 143.8, 133.3, 86.1, 73.1, 68.7, 33.1, 29.4, 22.7, 13.7; IR (neat) ν : 3418, 2113, 1693, 1128 cm^{-1} ; HRMS (CI) calcd for $\text{C}_{10}\text{H}_{15}\text{O}_2$ ($\text{M}^+ + \text{H}$) 167.1072, found

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167.1074.

(Z)-1-(4-Bromophenyl)-2-(1-(4-bromophenyl)-1-hydroxyprop-2-ynyl)pent-2-en-1-one 2h: The desired product was obtained according to the general procedure without addition of CH_2I_2 ; A yellow oil (80% yield); ^1H NMR (CDCl_3 , 300 MHz) δ : 7.60 (d, $J=8.4$ Hz, 2H), 7.49 (d, $J=8.4$ Hz, 2H), 7.45–7.38 (m, 4H), 5.99 (t, $J=6.9$ Hz, 1H), 4.38 (s, 1H), 2.66 (s, 1H), 1.76–1.66 (m, 2H), 0.79 (t, $J=7.5$ Hz, 3H); ^{13}C NMR (CDCl_3 , 75 MHz) δ : 198.8, 140.7, 139.5, 137.5, 136.5, 132.1, 131.4, 131.0, 129.4, 128.3, 122.4, 84.5, 76.2, 74.3, 23.6, 13.3; IR (neat) ν : 3442, 2113, 1666, 1232, 1173 cm^{-1} ; HRMS (EI) calcd for $\text{C}_{20}\text{H}_{16}\text{O}_2\text{Br}_2$ (M^+) 445.9512, found 445.9517.

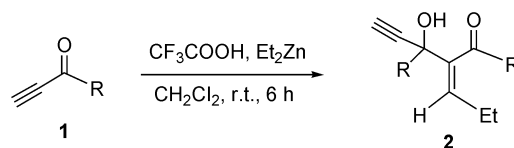
2,5-Dihydro-2-ethynyl-2-(p-chlorophenyl)-3-(p-chlorobenzoyl)furan 3c: The desired product was obtained according to the general procedure; A yellow oil (46% yield); ^1H NMR (CDCl_3 , 300 MHz) δ : 7.61–7.54 (m, 4H), 7.34 (d, $J=8.7$ Hz, 2H), 7.25 (d, $J=8.7$ Hz, 2H), 6.69 (t, $J=1.7$ Hz, 1H), 5.01 (d, $J=1.7$ Hz, 2H), 2.86 (s, 1H); ^{13}C NMR (CDCl_3 , 75 MHz) δ : 188.1, 142.4, 140.3, 139.6, 138.7, 136.4, 134.7, 130.4, 129.0, 128.5, 128.1, 86.7, 81.6, 76.6, 74.4; IR (neat) ν : 3293, 2110, 1650, 1242, 1091 cm^{-1} ; HRMS (EI) calcd for $\text{C}_{19}\text{H}_{12}\text{O}_2\text{Cl}_2$ (M^+) 342.0208, found 342.0214.

Results and discussion

Organozinc species $\text{CF}_3\text{CO}_2\text{ZnEt}$ can be readily prepared by stirring ZnEt_2 with CF_3COOH in CH_2Cl_2 at 0°C for 30 min. The *in situ* generated $\text{CF}_3\text{CO}_2\text{ZnEt}$ was treated with a variety of terminal acetylenic ketones, and the results were presented in Table 1. Both aliphatic and aromatic acetylenic ketones could be employed. For instance, treatment of 3-butyne-2-one with $\text{CF}_3\text{CO}_2\text{ZnEt}$ at room temperature for 6 h afforded β -substituted Morita-Baylis-Hillman adduct **2a** in a 72% isolated yield. The reaction of aromatic acetylenic ketones proceeded smoothly, providing the corresponding products in good

yields. The substituents on the benzene ring of aromatic acetylenic ketones seem to have no obvious effect on the yields.

Scheme 1

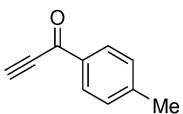
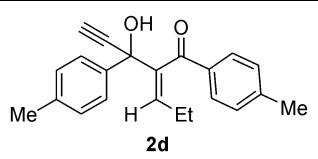
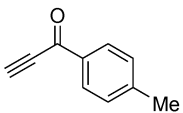
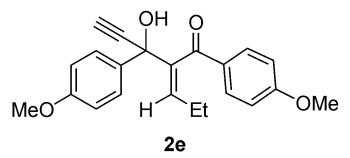
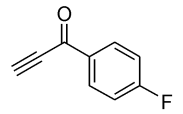
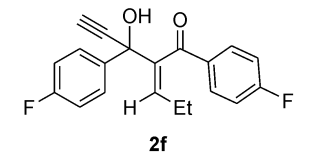
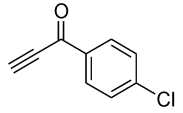
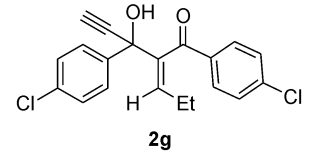
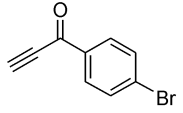
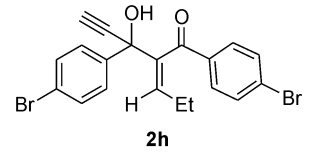
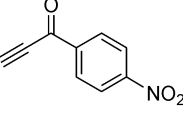
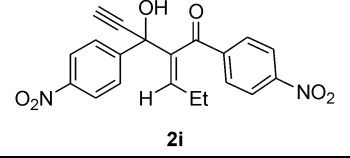


Organozinc species $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$, which is readily formed by the reaction of $\text{CF}_3\text{CO}_2\text{ZnEt}$ with CH_2I_2 at 0°C for 30 min, is an effective reagent for cyclopropanation of olefins.⁹ When $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$, in place of $\text{CF}_3\text{CO}_2\text{ZnEt}$, was used to this self-condensation reaction, the desired product **2** was not obtained. But a new compound was found, which was determined to be 2,5-dihydrofuran **3** by NMR and HRMS data. For example, stirring 1-phenylprop-2-yn-1-one with 1.0 equiv of $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$ at room temperature for 24 h afforded **3a** in a 40% yield. Some efforts have been made to improve the yield of **3**. Unfortunately, no better results were achieved by addition of CuI , increasing the reaction temperature, or lengthening the reaction time. The low yield of product **3** might be ascribed to the low reactivity of $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$ in the 1,4-Michael type addition to α,β -acetylenic ketones. At the same time, the reactivity of acetylenic ketones had a remarkable effect on the reaction yields. An electron-withdrawing group on the benzene ring of aromatic acetylenic ketone is beneficial to the formation of product **3**. If an electron-donating group was present on the benzene ring, the desired product was obtained in a lower yield. For example, a trace amount of product **3g** was observed when a methoxy group was on the benzene ring of aromatic acetylenic ketone. However, no reaction occurred using 3-butyne-2-one under our typical conditions due to the lower reactivity of alkylic acetylenic ketones than that of aromatic acetylenic ketones.

Table 1 Reactions of acetylenic ketones with organozinc species $\text{CF}_3\text{CO}_2\text{ZnEt}$

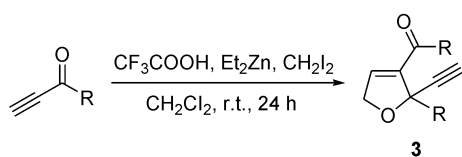
Entry	Substrate	Product	Yield ^a /%
1			72
2			70
3			89

Continued

Entry	Substrate	Product	Yield ^a /%
4		 2d	70
5		 2e	73
6		 2f	88
7		 2g	71
8		 2h	80
9		 2i	65

^a Yields after purification by column chromatography.

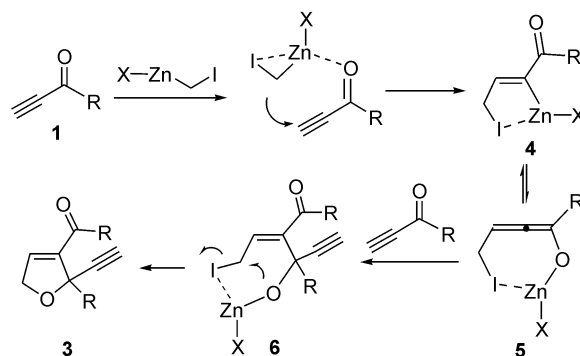
Scheme 2

**Table 2** Reactions of acetylenic ketones with organozinc species $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$

Entry	R	Product	Yield ^a /%
1	Ph	3a	40
2	4-FC ₆ H ₄	3b	30
3	4-ClC ₆ H ₄	3c	46
4	4-BrC ₆ H ₄	3d	47
5	4-NO ₂ C ₆ H ₄	3e	48
6	4-MeC ₆ H ₄	3f	32
7	4-MeOC ₆ H ₄	3g	<5
8	Me	NR ^b	

^a Yields after purification by column chromatography. ^b NR=no reaction.

A mechanism was postulated for the formation of product **3** (Scheme 1). The initial reaction step involves a reaction of organozinc species $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$ with acetylenic ketones through a Michael-type addition to give an active β -substituted zinc-allenolate intermediate **5**. The intermediate **5** attacks the carbonyl group of the

Scheme 3 Possible mechanism for the formation of 2,5-dihydrofuran

second acetylenic ketones to form **6**, followed by elimination of XZnI to provide product **3**.

In summary, a self-condensation reaction of α,β -acetylenic ketones via organozinc species $\text{CF}_3\text{CO}_2\text{ZnEt}$ has been developed. This reaction system provides functionalized trisubstituted olefins with good yields. We also found that $\text{CF}_3\text{CO}_2\text{ZnCH}_2\text{I}$ could react with active aromatic acetylenic ketones to provide product **3** in moderate yields.

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